

Telecom Customer Churn Analysis

End-to-End Analytics Project

Stage 3: Power BI — Interactive Dashboard

1. Stage 3 Overview

In Stage 3, all insights uncovered in Excel and SQL were brought to life through a 3-page interactive Power BI dashboard. The dashboard tells a complete business narrative — from what is happening, to why it is happening, to who is at risk and what it costs.

Dashboard Structure

Page 1: Executive Summary — What is happening? | Page 2: Deep Churn Dive — Why is it happening? | Page 3: Risk & Revenue — Who is at risk and what does it cost?

Tools & Setup

Tool: Power BI Desktop | Theme: Custom dark navy JSON theme | Canvas size: 1280 x 720 | Color palette: Teal #00C4B4 | Purple #7B6FDD | Red #FF6B6B | Amber #FFB347 | Background: #0A0E1A

2. DAX Measures

A total of 30+ DAX measures were created in a dedicated _Measures table to power all visuals across the 3 pages.

Foundation Measures

```
Total Customers = COUNTROWS(telecom_churn)

Total Churned =
COUNTROWS(FILTER(telecom_churn, telecom_churn[customer_status] = "Churned"))

Total Stayed =
COUNTROWS(FILTER(telecom_churn, telecom_churn[customer_status] = "Stayed"))

Total Joined =
COUNTROWS(FILTER(telecom_churn, telecom_churn[customer_status] = "Joined"))

Churn Rate = DIVIDE([Total Churned], [Total Customers], 0)
% Stayed = DIVIDE([Total Stayed], [Total Customers], 0)
% Joined = DIVIDE([Total Joined], [Total Customers], 0)
```

Revenue Measures

```
Total Revenue Lost =
CALCULATE(SUM(telecom_churn[total_revenue]),
```

```

    telecom_churn[customer_status] = "Churned")

Monthly Revenue Lost =
CALCULATE(SUM(telecom_churn[monthly_charge]),
    telecom_churn[customer_status] = "Churned")

Revenue at Risk Annual =
CALCULATE(SUM(telecom_churn[monthly_charge]),
    telecom_churn[customer_status] = "Churned") * 12

Total Revenue Stayed =
CALCULATE(SUM(telecom_churn[total_revenue]),
    telecom_churn[customer_status] = "Stayed")

Total Revenue Joined =
CALCULATE(SUM(telecom_churn[total_revenue]),
    telecom_churn[customer_status] = "Joined")

Revenue Lost to Churn =
CALCULATE(SUM(telecom_churn[total_revenue]),
    telecom_churn[customer_status] = "Churned")

```

Average Measures

```

Avg Monthly Charge = AVERAGE(telecom_churn[monthly_charge])

Avg Monthly Charge Churned =
CALCULATE(AVERAGE(telecom_churn[monthly_charge]),
    telecom_churn[customer_status] = "Churned")

Avg Monthly Charge Stayed =
CALCULATE(AVERAGE(telecom_churn[monthly_charge]),
    telecom_churn[customer_status] = "Stayed")

Avg Monthly Charge Joined =
CALCULATE(AVERAGE(telecom_churn[monthly_charge]),
    telecom_churn[customer_status] = "Joined")

Avg Tenure Churned =
CALCULATE(AVERAGE(telecom_churn[tenure_in_months]),
    telecom_churn[customer_status] = "Churned")

Avg Tenure Stayed =
CALCULATE(AVERAGE(telecom_churn[tenure_in_months]),
    telecom_churn[customer_status] = "Stayed")

Avg Tenure Joined =
CALCULATE(AVERAGE(telecom_churn[tenure_in_months]),
    telecom_churn[customer_status] = "Joined")

Loyal Customer Avg Revenue =
CALCULATE(AVERAGE(telecom_churn[total_revenue]),
    telecom_churn[customer_category] = "Loyal Customer")

```

Segment-Specific Measures

```
New Customer Churn Rate =
DIVIDE(
    CALCULATE([Total Churned], telecom_churn[customer_category] = "New Customer"),
    CALCULATE([Total Customers], telecom_churn[customer_category] = "New
Customer"), 0)

Fiber Optic Churn Rate =
DIVIDE(
    CALCULATE([Total Churned], telecom_churn[internet_type] = "Fiber Optic"),
    CALCULATE([Total Customers], telecom_churn[internet_type] = "Fiber Optic"), 0)

Offer E Churn Rate =
DIVIDE(
    CALCULATE([Total Churned], telecom_churn[offer] = "Offer E"),
    CALCULATE([Total Customers], telecom_churn[offer] = "Offer E"), 0)

San Diego Revenue Lost =
CALCULATE([Revenue Lost to Churn], telecom_churn[city] = "San Diego")

Temecula Revenue Lost =
CALCULATE([Revenue Lost to Churn], telecom_churn[city] = "Temecula")

Fallbrook Revenue Lost =
CALCULATE([Revenue Lost to Churn], telecom_churn[city] = "Fallbrook")
```

3. Page 1 — Executive Summary

Answers the question: What is happening? Designed for senior management — a quick snapshot of the overall churn situation with high level KPIs, customer segment cards and key pattern charts.

Row 1 — KPI Cards

Card	Measure	Format	Border Color
Total Customers	Total Customers	Whole Number	#7B6FDD
Churn Rate	Churn Rate	Percentage	#FF6B6B
Total Revenue Lost	Total Revenue Lost	Currency	#FFB347
Monthly Revenue Lost	Monthly Revenue Lost	Currency	#00C4B4

Row 2 — Customer Status Highlight Cards

Three large cards showing detailed metrics for each customer status group — Churned, Stayed and Joined. Each card contains 5 metrics stacked vertically.

Metric	Churned Card	Stayed Card	Joined Card
Count	Total Churned	Total Stayed	Total Joined
Rate	Churn Rate	% Stayed	% Joined
Total Revenue	Revenue Lost to Churn	Total Revenue Stayed	Total Revenue Joined
Avg Monthly Charge	Avg Monthly Charge Churned	Avg Monthly Charge Stayed	Avg Monthly Charge Joined

Avg Tenure	Avg Tenure Churned	Avg Tenure Stayed	Avg Tenure Joined
Border Color	#FF6B6B	#FFB347	#00C4B4

Row 3 — Charts

Visual	Type	Y-Axis	X-Axis	Color
Churned by Contract Type	Clustered Bar	contract	Total Churned	#00C4B4
Churned by Age Group	Clustered Bar	age_group	Total Churned	#7B6FDD

Row 4 — Detail Table

A full-width table showing churn breakdown by Contract Type + Internet Type combination with conditional formatting on the Total Revenue Lost column.

Column	Type	Format
Contract	Column	Text
Internet Type	Column	Text
Total Customers	Measure	Whole Number
Total Churned	Measure	Whole Number
Churn Rate	Measure	Percentage
Avg Monthly Charge	Measure	Currency
Total Revenue Lost	Measure	Currency — conditional red formatting

Page 1 Key Visuals

The Executive Summary page immediately communicates scale — 26.54% churn rate, \$3.68M revenue lost, and Month-to-Month Fiber Optic customers as the highest risk segment visible in the detail table.

4. Page 2 — Deep Churn Dive

Answers the question: Why is it happening? An analytical deep dive into the root causes of churn across multiple dimensions.

Row 1 — KPI Cards

Card	Measure/Value	Border Color
Top Churn Reason	Competitor had better devices — 313 customers	#FF6B6B
Top Churn by Internet Type	Fiber Optic — 1,236 of 3,035 customers	#7B6FDD
Top Churn Offer	Offer E — 426 of 805 customers	#00C4B4

Row 2 — Large Charts

Visual	Type	Setup	Filter
Top Churn Reasons	HTML Content Visual	Custom DAX HTML measure with progress bars	customer_status = Churned
Churn by Internet Type	Clustered Bar	Y: internet_type X: Churn Rate	Conditional color by churn rate

Row 3 — Smaller Charts

Visual	Type	Setup
Churn by Payment Method	Clustered Bar	Y: payment_method X: Churn Rate
Churn by Offer	Clustered Bar	Y: offer X: Churn Rate Sort descending
Churn by Tenure Group	HTML Content Visual	Custom DAX HTML measure with progress bars

Page 2 Key Visuals

The Deep Churn Dive page reveals that competitors account for 44.99% of all churn reasons, Fiber Optic customers churn at 40.72%, Offer E drives 52.92% churn while Offer A retains customers at just 6.73% churn, and add-on services reduce churn by up to 65%.

5. Page 3 — Risk & Revenue

Answers the question: Who is at risk and what does it cost? The strategic layer showing which customer segments are most at risk and the financial impact.

Row 1 — KPI Cards

Card	Measure	Format	Border Color
Revenue at Risk Annual	Revenue at Risk Annual	Currency	#FF6B6B
New Customer Churn Rate	New Customer Churn Rate	Percentage	#FFB347
Loyal Customer Avg Revenue	Loyal Customer Avg Revenue	Currency	#00C4B4
Avg Tenure Churned	Avg Tenure Churned	Whole Number	#7B6FDD

Row 2 — Revenue Lost by City + Tenure Group

Visual	Type	Setup
Revenue Lost by City	Treemap	Category: city Values: Revenue Lost to Churn Top N filter: 10 cities Conditional color by revenue
Churn Rate by Tenure Group	HTML Content Visual	Custom DAX HTML measure with color-coded progress bars

Row 3 — Referrals + City Risk Cards

Visual	Type	Setup
Churn Rate by Referrals	HTML Content Visual	Custom DAX HTML measure — 0 to 10 referrals with color-coded bars
Top 3 City Risk Cards	HTML Content Visual	Custom DAX HTML measure — San Diego, Temecula, Fallbrook with churn rate + revenue lost

HTML Content Measures Used

Four custom HTML DAX measures were created to produce styled progress bar visuals that match the dark theme:

Measure Name	Visual	Page
Top Churn Reasons HTML	Top 6 churn reasons with color-coded bars	Page 2
Tenure Churn HTML	Churn rate by tenure group with progress bars	Page 2 & 3
Referrals Churn HTML	Churn rate by 0-10 referrals with progress bars	Page 3
City Risk Cards HTML	San Diego, Temecula, Fallbrook risk cards	Page 3

Page 3 Key Visuals

The Risk & Revenue page reveals \$1.65M annual revenue at risk, new customers churn at 47.44% vs loyal customers at 9.51%, San Diego has lost \$385K in revenue with a 64.91% churn rate, and customers with 10 referrals have 0% churn.

6. Dashboard Theme & Design

Color Palette

Background: #0A0E1A | Card background: #131929 | Teal: #00C4B4 | Purple: #7B6FDD | Red: #FF6B6B | Amber: #FFB347 | Dark navy: #0A0E1A | Grid lines: #1e2535

Design Principles

Dark navy background for professional appearance | Color-coded borders by insight category | HTML Content visuals for custom progress bars | Consistent typography: Segoe UI throughout | All visuals have rounded corners and matching border colors

7. Stage 3 Summary

Page	Question Answered	Key Visual	Key Finding
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Page 1 — Executive Summary	What is happening?	4 KPI cards + 3 status cards + 2 charts + table	26.54% churn — \$3.68M revenue lost
Page 2 — Deep Churn Dive	Why is it happening?	HTML churn reasons + internet type + offer + tenure	Competitors 45% of churn — Offer E worst at 52.92%
Page 3 — Risk & Revenue	Who is at risk?	Treemap + HTML referrals + city risk cards	San Diego \$385K lost — New customers 47.44% churn

Project Complete

All 3 stages complete: Excel preprocessing + EDA | SQL deep analysis (13 queries) | Power BI 3-page interactive dashboard | 13 business insights | 10+ actionable recommendations | Full documentation for GitHub portfolio